

University of Wollongong
School of Computing and Information Technology
CSIT121 Object Oriented Design and Programming
Assignment 3

Objectives

- To apply Object Oriented Design (OOD).
- To apply Object Oriented Programming (OOP) using Python.

Note: Please read the submission requirement carefully. Scripts will be used to grade and update your score to Moodle. Failure to name your file name correctly in the required format will result in zero marks updated to Moodle.

Tasks

You are given a CSV file MovieSales.csv that contains movie sales record from the year 1990 onwards. Your task is to implement an Analytic class to read, store, and filter this data with the following methods:

Method	Description
<code>__init__(...)</code>	Pass in the name of the csv file. Load the CSV file on initialisation.
<code>get_genres()</code>	Return the list of unique genres.
<code>get_directors()</code>	Return the list of unique directors.
<code>count()</code>	Return the number of valid records read from the CSV file. Implement as a <code>@property</code> .
<code>match(...)</code>	Return a list of all matching records as dictionaries. Method definition: <pre>def match(self, title=[], genre=[], year_of_release=[], director=[], studio=[], global_sales=[], critic_score=[], rating=[])</pre> All are required arguments. For numeric fields like <code>year_of_release</code>, <code>global_sales</code>, and <code>critic_score</code>, pass in a list of two values as lower and upper bounds. <code>critic_score</code> is a value from 0 to 100 If more than one attribute is passed in, all must match (AND logic). If there is no match, pass back an empty list.

	Explanation for each attribute: <u>title</u> • pass in a list of movie title to match • for example, below will retrieve two movies with name “Mission Impossible and “The last kingdom” • name=[“Mission Impossible”, “The last kingdom”] For genre, director, studio and rating, the logic is the same as title. <u>year_of_release</u> • pass in two variables as lower and upper limit. • for example, below will retrieve all movies that were released from year 2010 to 2012. • year_of_release=[2010,2012] For global_sales and critic_score, the logic is the same as year_of_release. If more than one attributes are pass in, all the attributes must match (i.e. AND operation) For example: <pre>match(studio = [“Warner Bros”, “Universal”], year_of_release=[2010,2012])</pre> The above will return a list of movies that studio is either “Warner Bros” or “Universal” AND the year of release is from 2010 to 2012.
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Skip those records with errors and print the data lines with errors description to errors.txt file. The errors.txt should contain the line number and the error description. (see error.txt as a reference).

The error.txt is generated based on MovieSales_Errors.csv.
Analyse the sample error.txt to work out the data row error checking requirement.

Another test data MoviesSales_2.csv is also available for you to further test your code. It consists of realistic data for vigorous testing. You are strongly encouraged to try out MoviesSales_2.csv and work out test cases on your own to test your code.

Test Code and automated grading

The testing code StudentAssertionTest.py has been provided to assist you to test your code.

Ensure that you have passed all the test cases before submission, else zero marks will be awarded if the StudentAssertionTest.py failed to run.

Another script with comprehensive test cases will be used to automate the grading of your assignment.

Submission requirements (very important)

- Submit the python code in a txt file containing the code for all the classes. Do not include your test code.
- For full time student:
 - File name must be in the form of: **TXXF_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (**not SIM student ID number**).
 - For example, **T02F_JeffreyTan_8080426.txt**
- For part time student:
 - File name must be in the form of: **TXXP_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (**not SIM student ID number**).
 - For example, **T02P_JeffreyTan_8080426.txt**
- **Note**
 - Automated script will be used to extract your file for grading. Failure to follow the above instructions will result in failure to extract your submission correctly.
 - Ensure that the UOW ID number is correct as it will be used by the scripts to update your marks in Moodle. Wrong UOW ID number will result in your marks not updated in Moodle.
 - Note as there are many tutorial groups, failure to name your file name correctly in the above format might result in zero marks.
- Late submission will be penalized 5% per day late.
- Late submission more than 4 days late will be awarded a mark of zero

The End